

Lock-In Amplifier

PCB Design Using STM32H743

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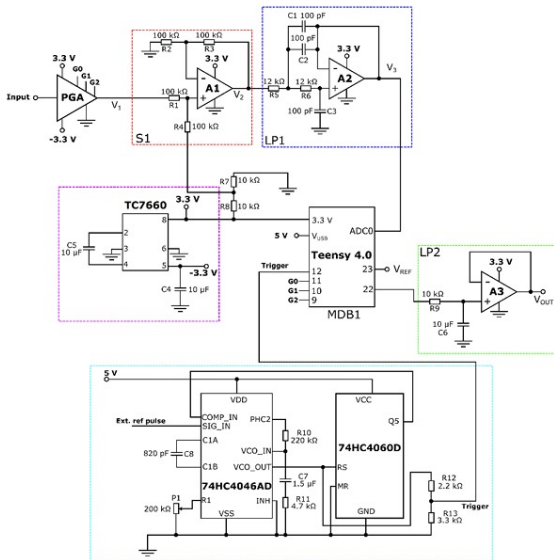
Motivation

- Many scientific measurements involve detecting extremely small signals.
- These signals are usually buried under large noise sources.
- Traditional measurement instruments cannot isolate such weak signals.
- Lock-in amplifiers solve this problem using synchronous detection.
- Commercial lock-in amplifiers are expensive.

Goal of this project

- Build a low-cost lock-in amplifier.
- Implement both analog front-end and digital signal processing.
- Design a custom PCB for high performance measurements.

Reference Design: OpenLockin



- Initial architecture based on OpenLockin open-source design.
- Original implementation used Teensy microcontroller.
- Includes analog signal conditioning and digital demodulation.
- Provides reference waveform generation and measurement output.

Our Design Improvements

- Replaced Teensy controller with STM32H743.
- STM32 provides 480 MHz processing capability.
- Built-in ADC and DAC peripherals simplify system design.
- Custom 4-layer PCB designed for mixed signal operation.
- Improved analog front-end for higher sensitivity.

Advantages:

- Higher processing speed
- Integrated DSP peripherals
- Better system scalability

What is a Lock-In Amplifier

- A lock-in amplifier extracts signals at a specific frequency.
- It rejects noise outside the reference frequency.
- Uses synchronous detection technique.
- Allows measurement of signals smaller than noise level.

Applications:

- Optical sensing
- Magnetic measurements
- Biomedical instruments
- Scientific experiments

Principle of Lock-In Detection

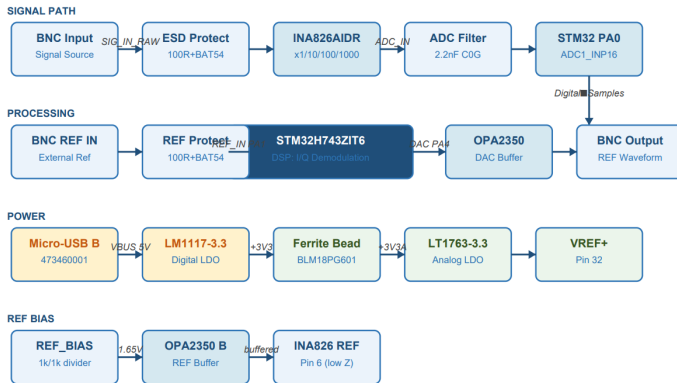
- Input signal contains desired signal and noise.
- The signal is multiplied with a reference signal.
- Desired signal moves to DC after multiplication.
- Low pass filter removes high frequency components.

$$V_{sig} = A_{sig} \cos(\omega t + \theta)$$

$$V_{ref} = A_{ref} \cos(\omega t)$$

$$Output = \frac{A_{sig} A_{ref}}{2} \cos(\theta)$$

System Block Diagram



Signal flow:

- Sensor signal enters through BNC connector
- Signal amplified using instrumentation amplifier
- STM32 samples signal using ADC
- Digital lock-in algorithm extracts amplitude and phase

Signal Flow in the System

- 1 Sensor signal enters input stage.
- 2 Instrumentation amplifier increases signal amplitude.
- 3 Anti-aliasing filter removes high frequency noise.
- 4 STM32 ADC samples signal.
- 5 Reference sine wave generated by DAC.
- 6 Digital multiplication extracts signal component.

STM32H743 Microcontroller

- 480 MHz ARM Cortex-M7 processor
- Hardware floating point unit
- 12-bit ADC for signal acquisition
- 12-bit DAC for reference generation
- DMA support for fast data transfer

Role:

- Signal acquisition
- Digital demodulation
- Reference signal generation
- Data communication with PC

Instrumentation amplifier:

INA826

- Amplifies weak signals before ADC sampling
- Gain selectable from x1 to x1000
- High common-mode rejection ratio
- Low noise operation

Additional features:

- REF pin sets output midpoint
- Input protection using Schottky diodes
- Anti-aliasing filter before ADC

Reference Signal Generation

- Reference sine wave generated using STM32 DAC.
- DAC driven using lookup table and DMA.
- Output buffered using OPA2350 op-amp.
- Provides stable reference waveform.

Advantages:

- Clean analog sine wave
- Wide frequency range
- Avoids PWM filtering limitations

Precision analog measurements require low noise power supply.

Design uses two regulators:

- LM1117 for digital circuits
- LT1763 low-noise regulator for analog circuits

Additional features:

- Ferrite bead isolation
- Separate analog ground
- Star-ground connection

Digital Lock-In Algorithm

Steps implemented in firmware:

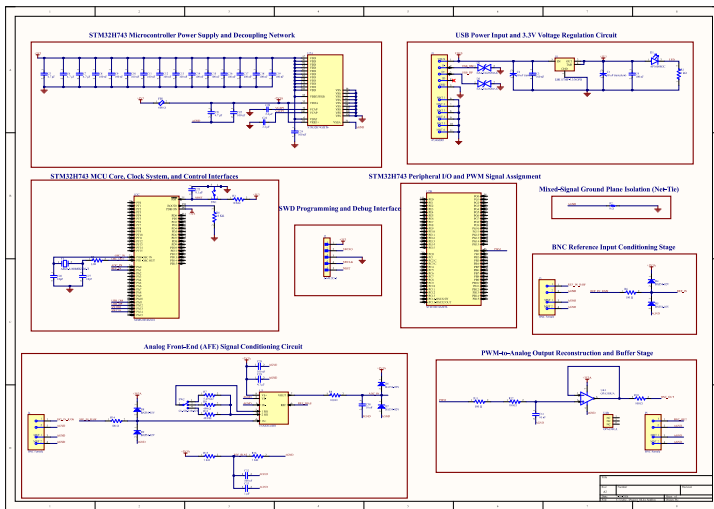
- 1 Generate reference sine wave.
- 2 Sample signal using ADC.
- 3 Multiply with sine and cosine reference.
- 4 Compute in-phase and quadrature components.
- 5 Calculate amplitude and phase.

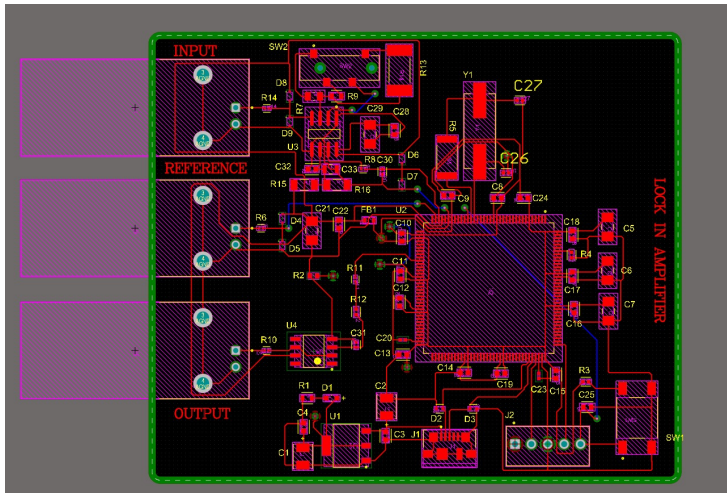
$$R = \sqrt{I^2 + Q^2}$$

$$\theta = \tan^{-1}(Q/I)$$

Key PCB features:

- 4-layer PCB stackup
- Separate analog and digital ground planes
- Short analog signal paths
- Proper decoupling capacitors
- Controlled impedance USB routing





Expected Performance

- Frequency range: 0.1 Hz – 100 kHz
- Gain range: x1 to x1000
- ADC resolution: 12-bit
- Sampling rate up to 200 kSPS
- Microvolt-level signal detection capability
- USB powered system

Future Work

- Fabrication of the PCB.
- Hardware testing and validation.
- Calibration of analog front-end.
- Implementation of improved digital filtering.
- Development of PC interface for visualization.

Conclusion

- Designed a digital lock-in amplifier architecture.
- Integrated analog amplification with STM32 DSP processing.
- Developed a custom mixed-signal PCB design.
- Provides a low-cost alternative to commercial systems.